

Statistics 6910, Section 004, Fall 2013, Quiz 2 (25 Points)

Friday, November 22, 2013

Your Name: _____

Question 1: Reading Regular Expressions (14 Points)

Assume that we have the following text strings:

```
> exprTest = c("abcde", "aa-bbccdd", "baba", "abba", # 1 - 4
+             "caab", "ab-ba", "ebba", "bea" # 5 - 8
+             )
```

For each of the following regular expressions, indicate the number of the text string(s) that are matched. There may be 0 to 8 matches in each case. If the answer is 0, write “no match”. One example is given below:

```
> grep("aa", exprTest)
```

```
[1] 2 5
```

1. `> grep("[ce][ea]", exprTest)`

2. `> grep("-c", exprTest)`

3. `> grep("[-ce]", exprTest)`

4. `> grep("[c-e]", exprTest)`

5. `> grep("a.*a", exprTest)`

6. `> grep("a[~a]+a", exprTest)`

7. `> grep("^a.*a$", exprTest)`

Question 2: Writing Regular Expressions (11 Points)

1. (5 Points) Write a regular expression that matches texts with a weekday from Monday through Friday. However, if only Saturday and Sunday are part of the text (without a weekday from Monday through Friday), this won't be a match. All valid days start with an uppercase letter.

Assume that the following test strings are used:

```
> dayTest = c("Friday on my mind.", "I love Sundays",  
+             "I teach on Mondays!", "Friday and Saturday",  
+             "Everyday is like Sunday.")
```

The return value should be:

```
[1] 1 3 4
```

A possible regular expression for this task is given below:

2. (6 Points) Write a regular expression that matches a simplified full US telephone number, with or without a leading 1. Only spaces are allowed as part of the number. The formatting must be 1 plus space (optional), followed by 3 digits, space, 3 digits, space, and 4 digits. The number must be at the start of the text. Parentheses and “-” are not allowed as part of the number. We must have the proper spacing as indicated above!

Note that there are some typos in the lecture notes. It should correctly read:

`{n}`

The preceding item is matched exactly `n` times.

`{n,}`

The preceding item is matched `n` or more times.

`{n,m}`

The preceding item is matched at least `n` times, but not more than `m` times.

Assume that the following test strings are used:

```
> numberTest = c("1 435 797 0696", "435 797 0696", "1-435-797-0696",  
+               "435-797-0696", "(435) 797 0696", "1 (435) 797 0696",  
+               "14357970696", "435 7970 696", "1 435 797 069",  
+               "Call 1 435 797 0696", "9 435 797 0696",  
+               "1435 797 0696", " 435 797 0696")
```

The return value should be:

```
[1] 1 2
```

To make spaces visible in your regular expression, use the symbol `\` to clearly mark each space in your expression!

A possible regular expression for this task is given below: